

Scoped Obstructions to Stationarizing a Graded Wheel-Sieve Symbolic System

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Abstract

A graded wheel-sieve symbolic system advances through disjoint levels while generating the exact clock increment $\tau_k = \log q_{k+1}$ from its consecutive-prime recursion. We ask which stationary symbolic transformations could add intrinsic periodic orbits without erasing that endogenous clock. Three scoped obstructions screen the most immediate constructions. First, any dynamical system semiconjugating to the graded shift has no periodic points, and the standard inverse-limit natural extension is empty. Second, a strong forward-bisimulation quotient of a finite directed acyclic graph is acyclic; independently, a quotient that keeps the exact next multiplier as a state-class label inherits the strict level grading. Third, a fixed finite-window decoder over a finite alphabet has finite image and therefore cannot recover the unbounded sequence of exact prime multipliers. These statements are deliberately not a universal no-go theorem. The finite-graph result does not cover infinite directed acyclic graphs, and the decoder result does not cover countable alphabets or infinite memory. The only surviving class considered here is thus a genuinely new infinite factor or observational recoding with exact decoders and a compatible path-lifting rule. That object has not been source-locked, so it remains *not testable*. This stage contains no numerical experiment, assigns no new candidate, defines no determinant, and leaves the analytic route locked.

Research status. This is a theorem-only stage paper. The surviving observational-recoding branch is NOT TESTABLE because its infinite source lock is incomplete. No Stage-02 numerical experiment has been run, no candidate SD-C07 has been assigned, no determinant has been defined, and Route B remains locked.

1 Introduction

A symbolic system can acquire periodic words when one forgets enough of its state. That observation alone is not useful for an arithmetic symbolic model: the same forgetting can destroy the mechanism that gave the symbols their arithmetic meaning. The wheel-sieve system studied here makes this tension explicit. Its shift advances through a sequence of disjoint levels, and its level transition generates the next prime multiplier. The grading prevents periodic points, while the distinct, unbounded multipliers provide the exact clock that a proposed deformation must retain.

The central question is therefore categorical rather than computational: which kind of symbolic transformation is being proposed? A strict extension must still project to the original shift. A strong-bisimulation quotient must match successor behavior between equivalent states. A local recoding must expose the clock through a fixed observation rule. These three requirements lead to different theorems, with different scope boundaries. Treating them as interchangeable would either overstate a negative result or credit a cycle that no longer carries the endogenous arithmetic.

This note gives a theorem-level screen before any finite-cutoff search. Its one-sentence conclusion is:

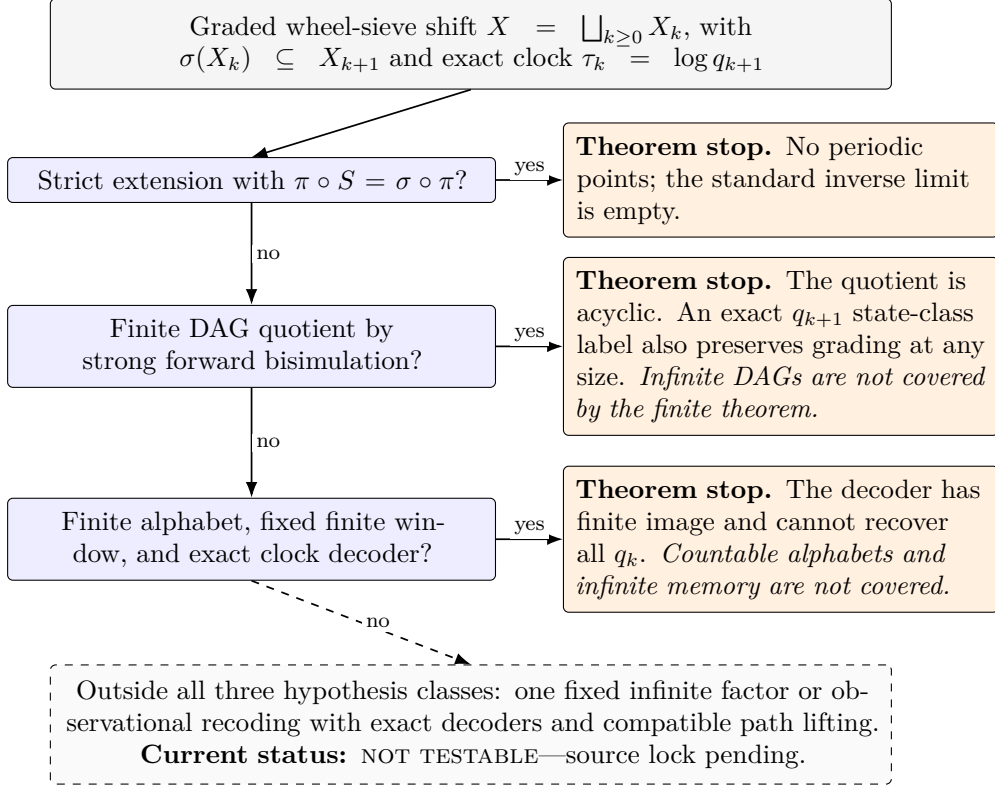


Figure 1: The obstruction map is hypothesis-sensitive. Strict semiconjugate extensions cannot gain periodic points; finite strong-bisimulation quotients of DAGs remain acyclic, while that theorem does not cover infinite DAGs; and finite-alphabet fixed-window recodings cannot decode the unbounded exact clock, while countable alphabets and infinite memory are not covered. The remaining infinite observational-recoding class is unresolved rather than validated: its source lock is incomplete and it is not testable.

Strict extensions, finite strong-bisimulation quotients, and finite-local exact-clock recodings are obstructed, but an explicitly defined infinite observational recoding remains outside those theorem classes and has not yet become a testable object.

The contributions are precise and falsifiable.

- C1.** We prove that an equivariant extension of a strictly graded shift has no periodic points. We also prove that the usual inverse-limit natural extension is empty because arbitrarily long backward level histories do not exist.
- C2.** We prove that a strong forward-bisimulation quotient of a *finite* directed acyclic graph is acyclic. We exhibit the infinite ray as a counterexample when finiteness is removed. A separate label-preservation proposition shows that an exact next-prime state label prevents cross-level merging even in an infinite graph.
- C3.** We prove that a fixed finite-window decoder over a finite alphabet cannot output the unbounded exact prime clock. The proof is a finite-image argument and therefore says nothing against countable alphabets or infinite-memory rules.

The diagram in Figure 1 is also a statement of what the paper does *not* prove. The finite-bisimulation theorem cannot be applied to an arbitrary infinite DAG. The local-decoder theorem cannot be applied to a countable alphabet or a rule with unbounded memory. Those escape classes do not automatically solve the problem: they must still generate the exact clock

internally and must distinguish genuine periodic paths from quotient cycles assembled from incompatible representatives.

The present evidence is entirely deductive. No Stage-02 numerical run has been executed, and finite cutoffs are not used as evidence for an infinite object. The putative observational recoding lacks a target phase space, a fixed rule, exact decoders, and path-lifting semantics. Its status is therefore NOT TESTABLE. In particular, this note does not introduce SD-C07, select a determinant, or unlock Route B.

The remainder of the paper defines the graded model and the relevant categories (Section 2), proves the strict-extension obstruction (Section 3), proves the quotient obstructions (Section 4), and isolates the clock-locality obstruction (Section 5). Section 6 specifies the obligations that a surviving infinite recoding would have to meet. The appendices audit the assumption boundaries and record the incomplete source-lock fields.

2 Graded Wheel Setup and Category Discipline

The proofs use only a small part of the wheel recursion. We isolate those properties so that every conclusion can be traced to its assumptions.

Definition 2.1 (Strictly graded shift). Let

$$X = \bigsqcup_{k \geq 0} X_k \quad (1)$$

be a disjoint union, and let $\sigma : X \rightarrow X$ satisfy

$$\sigma(X_k) \subseteq X_{k+1} \quad (k \geq 0). \quad (2)$$

We call (X, σ) a strictly graded shift and write $\text{lev}(x) = k$ for $x \in X_k$.

For the frozen wheel-sieve path system, level k carries an exact next multiplier q_{k+1} . The sequence $(q_k)_{k \geq 1}$ consists of distinct, unbounded rational primes generated by the same recursion, and the clock increment is

$$\tau_k = \log q_{k+1}. \quad (3)$$

No theorem below reads a prime table. The quotient result needs only that q_{k+1} is distinct across levels; the decoder result needs only that its range is infinite.

Definition 2.2 (Strict extension). A wheel-preserving strict extension is a dynamical system (Y, S) together with a map $\pi : Y \rightarrow X$ satisfying

$$\pi \circ S = \sigma \circ \pi. \quad (4)$$

Surjectivity of π , customary in the word *extension*, is not needed for the obstruction.

For graph quotients, let $G = (V, E)$ be a directed graph and let $\sim$ be an equivalence relation on V .

Definition 2.3 (Strong forward bisimulation). The relation $\sim$ is a strong forward bisimulation when, for all $v \sim w$,

- (i) if vEv' , then some w' satisfies wEw' and $v' \sim w'$; and
- (ii) if wEw' , then some v' satisfies vEv' and $v' \sim w'$.

The quotient graph $G/\sim$ has an edge $[v] \rightarrow [v']$ exactly when some representatives $u \in [v]$ and $u' \in [v']$ satisfy uEu' .

Table 1: Each obstruction is tied to an explicit hypothesis class. The rightmost column is part of the result, not a proposed construction.

Class	Conclusion	Not covered
Strict extension satisfying (4)	No periodic points; standard inverse limit empty	New factor, quotient, recoding, or independent grammar
Finite DAG plus strong forward bisimulation	Quotient acyclic	Infinite DAG; one-way simulation; arbitrary graph image
Level-injective state-class label	Quotient inherits strict grading	Edge or path decoder that distinguishes representatives
Finite alphabet plus fixed finite window	Exact unbounded clock undecodable	Countable alphabet or infinite memory

Finally, a finite-local clock rule consists of a finite alphabet A , a fixed finite window W , and one fixed decoder $d : A^W \rightarrow \mathbb{N}$. The words *fixed* and *finite* exclude a cutoff-dependent lookup table or a window that grows with the level.

The discipline in Table 1 prevents two errors. First, a theorem cannot be enlarged by silently dropping one of its hypotheses. Second, an object outside a no-go theorem does not receive positive evidence merely by escaping that theorem.

3 Strict Extensions Cannot Supply Periodic Orbits

The grading already closes the most literal stationarization proposal.

Theorem 3.1 (Periodic-point projection obstruction). *Let (X, σ) satisfy (1)–(2). If (Y, S) and $\pi : Y \rightarrow X$ satisfy (4), then*

$$\text{Fix}(S^n) = \emptyset \quad \text{for every } n \geq 1. \quad (5)$$

Proof. Suppose $S^n y = y$ for some $y \in Y$ and $n \geq 1$. Iterating (4) gives

$$\sigma^n(\pi(y)) = \pi(S^n y) = \pi(y).$$

If $\pi(y) \in X_k$, strict grading places the left-hand side in X_{k+n} and the right-hand side in X_k . The components are disjoint, so equality is impossible. $\square$

The standard natural-extension construction also collapses, for a different but related reason.

Proposition 3.2 (Empty standard inverse limit). *For a strictly graded shift,*

$$\varprojlim (X, \sigma) = \{(x_0, x_{-1}, \dots) : \sigma(x_{-j}) = x_{-j+1} \text{ for every } j \geq 1\} = \emptyset. \quad (6)$$

Equivalently,

$$\bigcap_{n \geq 0} \sigma^n(X) = \emptyset. \quad (7)$$

Proof. Assume $x_0 \in X_k$ has a full backward orbit. Because every application of σ raises the level by one, disjointness forces $x_{-j} \in X_{k-j}$. No component with negative index exists, contradicting the requirement when $j > k$.

For the equivalent formulation, (2) yields $\sigma^n(X) \subseteq \bigsqcup_{k \geq n} X_k$. A point has one finite level, so it cannot belong to all of these tails. $\square$

Remark 3.3 (Scope). Theorem 3.1 and proposition 3.2 concern systems that retain an equivariant map *to the frozen wheel shift*. A factor, observational recoding, quotient, or independently defined stationary grammar may not satisfy (4) in this direction. Such a construction is not refuted here, but neither is it an extension of the stated kind; it must re-establish its arithmetic interpretation from its own definition.

4 Strong-Bisimulation Quotients: Finite and Labelled Cases

A quotient may contain a cycle even when its source graph does not: different quotient edges can be witnessed by incompatible representatives. Strong forward bisimulation prevents that defect from creating a cycle in a *finite* DAG.

Theorem 4.1 (Finite-DAG strong-bisimulation obstruction). *Let $G = (V, E)$ be a finite directed acyclic graph and let $\sim$ be a strong forward bisimulation. Then the quotient graph $G/\sim$ is acyclic.*

Proof. Assume, for contradiction, that the quotient contains a directed cycle

$$C_0 \longrightarrow C_1 \longrightarrow \cdots \longrightarrow C_{m-1} \longrightarrow C_m = C_0, \quad m \geq 1.$$

For each $i \in \{0, \dots, m-1\}$, choose representatives $a_i \in C_i$ and $b_{i+1} \in C_{i+1}$ such that $a_i E b_{i+1}$, with indices read modulo m .

Choose any $v_0 \in C_0$. Suppose $v_j \in C_i$ has been chosen, where $i \equiv j \pmod{m}$. Since $v_j \sim a_i$ and $a_i E b_{i+1}$, successor matching supplies v_{j+1} such that

$$v_j E v_{j+1} \quad \text{and} \quad v_{j+1} \sim b_{i+1}.$$

Thus $v_{j+1} \in C_{i+1}$. Induction constructs an infinite directed path $v_0 E v_1 E v_2 E \cdots$ in G .

Because V is finite, two vertices on this path coincide. The positive path segment between two occurrences of the same vertex is a directed cycle in G , contradicting acyclicity. $\square$

The finiteness hypothesis is essential rather than technical.

Remark 4.2 (Infinite-ray counterexample). Let $V = \mathbb{N}$ and put one edge $n \rightarrow n+1$ at every vertex. This infinite graph is acyclic. The universal equivalence relation, under which all vertices are equivalent, is a strong forward bisimulation: the unique successors of any two vertices are again equivalent. Its quotient has one vertex and a self-loop. Therefore theorem 4.1 says nothing about arbitrary infinite DAGs.

For the wheel system, an independent grading argument applies when the exact clock value is required to be a quotient-state label.

Proposition 4.3 (Level-injective labels preserve acyclicity). *Let $V = \bigsqcup_{k \geq 0} V_k$, and suppose every edge goes from V_k to V_{k+1} . Let $\lambda : V \rightarrow \Lambda$ be constant on each V_k , with pairwise distinct values across levels. If*

$$v \sim w \implies \lambda(v) = \lambda(w), \tag{8}$$

then $G/\sim$ is acyclic. Graph finiteness and bisimulation are not needed.

Proof. Condition (8) and injectivity of the level labels force every equivalence class to lie in a single V_k . Define

$$\overline{\text{lev}}([v]) = k \quad \text{when } v \in V_k.$$

This quotient grading is well-defined. If $[v] \rightarrow [w]$, some representatives $v' \in [v]$ and $w' \in [w]$ satisfy $v' E w'$, hence

$$\overline{\text{lev}}([w]) = \overline{\text{lev}}([v]) + 1.$$

The quotient level increases along every positive-length path, so no path can return to its starting class. $\square$

Corollary 4.4 (Exact next-prime state labels). *If every class of a wheel-graph quotient carries one exact state-only value q_{k+1} for the next multiplier, then the quotient cannot merge different levels and cannot contain a directed cycle.*

Proof. At level k , set $\lambda(v) = q_{k+1}$. Consecutive rational primes are pairwise distinct, so proposition 4.3 applies. $\square$

Remark 4.5 (What the quotient results do not cover). The proof of theorem 4.1 uses the successor-matching property and source finiteness. It does not apply to a one-way simulation, a bounded-radius observational equivalence, or an arbitrary graph homomorphic image. The proof of proposition 4.3 applies only when the exact multiplier is constant on state classes. An edge decoder or a path-window decoder that distinguishes representatives within one state class falls outside that proposition. Falling outside either theorem creates an open proof obligation, not evidence of an arithmetic periodic orbit.

For a finite cutoff, theorem 4.1 removes the proposed positive branch entirely: a correctly computed strong-bisimulation quotient cannot produce a cycle. Running the quotient code could still serve as a regression test, but it could not supply positive Stage-02 evidence.

5 The Exact-Clock Locality Obstruction

Removing the explicit level label does not by itself make the arithmetic clock local. A fixed finite symbolic observation has only finitely many possible values.

Theorem 5.1 (Finite-local exact-clock obstruction). *Let A be a finite alphabet, let W be a fixed finite window, and let $d : A^W \rightarrow \mathbb{N}$ be a fixed decoder. Then d cannot output every exact wheel multiplier q_k . The same conclusion holds for a decoder whose required outputs are the exact values $\log q_k$.*

Proof. The pattern set A^W is finite because both A and W are finite. Therefore $\text{im}(d)$ is finite. The exact multipliers (q_k) form an infinite unbounded set, so they cannot all lie in $\text{im}(d)$. Since the logarithm is injective on positive numbers, the set $(\log q_k)$ is also infinite and cannot be the image of a finite domain. $\square$

Corollary 5.2 (Necessary escape from finite locality). *Any recoding that recovers all exact multipliers with one unchanged rule must drop at least one hypothesis of theorem 5.1: it must use a countable or larger alphabet, an unbounded observation domain or memory, or an additional unbounded input to the decoder.*

Corollary 5.2 is necessary, not sufficient. A countable alphabet can carry unbounded data, and an infinite-memory map can inspect unbounded context, so neither class is excluded by the finite-image proof. The theorem does not establish that either class admits a shift-commuting, level-blind, arithmetic-faithful recoding.

There is also a distinction between escaping the theorem and satisfying the research scope. A decoder that reads the external level, consults a stored prime table, or changes with cutoff has an infinite source of information and may evade the cardinality argument. Such a decoder does not recover the clock from one frozen endogenous grammar. Its failure is a source-lock or arithmetic-fidelity failure rather than a consequence of theorem 5.1.

The exact boundary is therefore:

Finite alphabet plus fixed finite window is impossible for exact unbounded clock recovery. Countable alphabet or infinite memory is unclassified until one explicit function space, topology, shift rule, and decoder are defined.

Table 2: Stage-02 decisions are consequences of theorem hypotheses or of missing definitions. None is a numerical verdict.

Proposed class	Decision	Reason
Strict semiconjugate extension	THEOREM STOP	Theorem 3.1 and proposition 3.2
Finite-DAG strong-bisimulation quotient expected to create cycles	THEOREM STOP	Theorem 4.1
Quotient with exact q_{k+1} as a state-class label	THEOREM STOP	Proposition 4.3 and corollary 4.4
Finite-alphabet fixed-window exact-clock recoding	THEOREM STOP	Theorem 5.1
Infinite observational recoding with countable alphabet or infinite memory	NOT TESTABLE	Target object, rule, decoders, and path lifting remain undefined
Determinant or analytic comparison	Locked	No infinite arithmetic and primitive-orbit ledger exists

6 The Surviving Class and Its Current Status

The preceding results leave one broad possibility: a new infinite factor or observational recoding that is neither a strict extension in the direction of (4), nor a finite strong-bisimulation quotient, nor a finite-local exact-clock code. The word *possibility* is logical. It does not name a construction and does not supply evidence.

For that branch to become a mathematical object, a source lock must specify at least the following data.

1. **Category and dynamics.** The target phase space Y , the target shift S , the map direction, the commutative relation, and whether Y is an image or an orbit closure.
2. **Grammar and regularity.** The alphabet and its topology if countable; one level-blind rule independent of cutoff; and either a coding radius or a named infinite-memory function class with its continuity requirements.
3. **Labels and transitions.** Vertex and edge labels, treatment of parallel edges and deleted branches, and a target transition rule that is fixed on the infinite system.
4. **Exact decoders.** Total arithmetic and clock decoders, with proofs that they recover the unit-residue data and the exact identity $\tau_k = \log q_{k+1}$ without an external level, prime table, or cutoff-dependent rule.
5. **Path compatibility.** A closed target word must lift to one compatible source path. It is insufficient for each target edge to have some source representative if adjacent edges require mutually incompatible representatives.
6. **Finite approximation.** Terminal markers, an interior comparison region, and restriction or extension maps that make every cutoff an approximation to the same infinite object.

Path compatibility is especially important. Suppose a quotient has edges $C_i \rightarrow C_{i+1}$ and each edge has a source witness. Without a lifting rule, the terminal representative of the witness for $C_i \rightarrow C_{i+1}$ need not be the initial representative of any witness for $C_{i+1} \rightarrow C_{i+2}$. A closed word in the coarse graph can therefore be assembled from locally valid but globally incompatible edges. Such a word is not an intrinsic periodic orbit of the arithmetic source.

The current source-lock document leaves the target system, alphabet, rule, decoders, closure convention, and finite-approximation semantics unresolved. Accordingly, no implementation run is authorized. Even a later finite periodic witness would authorize only an infinite theorem

attempt; it would not by itself establish arithmetic fidelity or completeness of a primitive orbit ledger.

The governance consequence is unambiguous. The surviving class remains NOT TESTABLE. There is no Stage-02 numerical evidence, no candidate SD-C07, and no determinant convention. Route B remains locked.

7 Conclusion and Limitations

Three scoped obstructions now delimit the wheel-sieve stationarization problem. Equivariant strict extensions cannot contain periodic points, and the standard inverse limit is empty. Strong forward-bisimulation cannot create cycles from a finite DAG; if the exact next prime is a state-class label, a separate grading argument blocks cross-level merging without a finiteness assumption. A finite alphabet observed through one fixed finite window cannot decode the unbounded exact clock.

These results close natural but narrow construction classes. They do not form a universal no-go theorem. An infinite DAG may have a cyclic strong- bisimulation quotient, as the infinite-ray example shows. A countable alphabet or an infinite-memory decoder falls outside the finite-image argument. Neither escape, however, supplies the missing arithmetic decoder, shift-commuting rule, or compatible path lift.

The next research step is therefore definitional rather than empirical: lock one infinite observational recoding and prove that it is a single, cutoff-independent symbolic system. Until that task is complete, the branch is NOT TESTABLE. This stage supports no numerical claim, assigns no SD-C07, and defines no determinant; Route B stays locked.

A Proof-Dependency and Boundary Audit

This appendix records exactly where each assumption enters. It is intended to prevent later finite experiments or recoding proposals from silently enlarging the theorems.

Table 3: Assumption audit for the proved statements.

Result	Assumptions actually used	Failure or escape when removed
Theorem 3.1	Disjoint levels, strict level growth, equivariance	No conclusion for a new object without a map to the frozen shift
Proposition 3.2	Nonnegative levels and one-level forward growth	A bi-infinite grading or added predecessors changes the object
Theorem 4.1	Finite source, source acyclic, successor matching	Infinite ray gives a cyclic quotient; weaker quotients need not lift paths
Proposition 4.3	One-level edge growth and a class label injective in level	No conclusion for edge/path decoders that vary within a state class
Theorem 5.1	Finite alphabet, fixed finite window, fixed decoder, infinite exact output set	Countable alphabet or infinite memory avoids the finite-domain step

A.1 The infinite-ray boundary in full

Let G have vertices v_n for $n \in \mathbb{N}$ and edges $v_n \rightarrow v_{n+1}$. A directed cycle would require the index to return to a smaller value, so G is acyclic. Put $v_n \sim v_m$ for all n, m . Given any related vertices, their unique successors are again related, in both directions required by strong forward bisimulation. The quotient has one class C , and every source edge witnesses $C \rightarrow C$. Thus $G/\sim$ has a self-loop. This example blocks any attempt to delete the word *finite* from theorem 4.1.

The counterexample does not validate an infinite wheel quotient. Its one-class quotient has forgotten every level-dependent value, including the exact wheel multiplier. It demonstrates a theorem boundary and nothing more.

A.2 Why cutoff stability is not an infinite definition

Suppose partitions are computed independently at several finite cutoffs. Even identical-looking class counts do not specify an infinite quotient. A definition still needs restriction or extension maps that commute with the source and target shifts, a treatment of terminal classes, and a proof that interior classes are consistent as the cutoff changes. Without those data, one cannot decide whether a finite closed word persists, whether it is a boundary artifact, or whether its edge witnesses lift compatibly in the infinite source.

The preregistered cutoffs can test an already defined infinite rule. They cannot supply the rule, its decoder, or its consistency theorem.

B Source-Lock Readiness Ledger

The unresolved branch has no candidate identifier. The following ledger records the fields that must be frozen before code can provide relevant evidence.

Field	Required content	Current state
Category and maps	Exactly one of factor, quotient, or recoding; target space and shift; map direction; commutative relation; image/closure convention	Pending
Alphabet and regularity	Alphabet topology if countable; coding radius or named infinite-memory class; continuity and complexity bounds	Pending
Frozen rule	One level-blind infinite rule, canonical serialization, and proof of cutoff independence	Pending
Labels and transitions	Vertex/edge labels, parallel-edge multiplicity, deleted-branch convention, and target transition rule	Pending
Arithmetic decoder	Total domain and formula; exact unit-residue recovery without a prime table or external level	Pending
Clock decoder	State/edge/window domain and exact recovery of q_{k+1} and $\log q_{k+1}$	Pending
Path lifting	Representative compatibility for every claimed closed word	Pending
Finite approximation	Terminal marker, interior region, and restriction/extension consistency equation	Pending
Analytic layer	Function space and determinant convention, only after an infinite arithmetic and orbit ledger	Locked

Because all defining fields remain pending, the observational-recoding branch is NOT TESTABLE. The correct next action is to complete one infinite source lock, not to search across finite quotient rules or enlarge a cutoff after seeing results.

B.1 Evidence and sharing record

This manuscript uses only the Stage-02 proof packages for the strict- extension, bisimulation, grading, and local-decoder arguments. The proofs have been restated in full so that the PDF is mathematically self-contained. The theorem relationship figure is pure TikZ and contains no measurements. No external citation metadata, experimental result, random seed, or compute claim enters the paper.

The sharing status at this draft is:

- anonymous author block;
- theorem-only evidence;
- no Stage-02 numerical run;
- no SD-C07;
- determinant convention not defined; and
- Route B locked.